

Week 2 - Assignment 2

Q. Two events having non-zero probabilities
— cannot be both mutually exclusive & independent

solⁿ The 2 events are mutually exclusive if they cannot occur at the same time.

If the occurrence of one event does not affect the occurrence of other event, the 2 events are independent. Therefore 2 events cannot occur at the same time.

Let suppose A & B events are mutually exclusive the probability of occurring at the same time is zero.

$$P(A \cap B) = 0$$

If these 2 events are independent of each other, the probability that both occur at the same time is,

$$P(A \cap B) = P(A) \cdot P(B).$$

if either $P(A)$ or $P(B)$ is zero,
 $P(A \cap B) = 0$.

In this case, the events can be considered not independent & mutually exclusive at the same time.

Two independent events cannot be mutually exclusive unless the probability of one or both events is zero. The occurrence of A affects the occurrence of B if the 2 events are mutually exclusive

If A & B are both independent & mutually exclusive
 $P(A \cap B) = P(A) \cdot P(B)$ — if A & B are independent
 $P(A \cap B) = 0$ — if A & B are mutually exclusive
Combining these 2 eqⁿ

$$P(A) \cdot P(B) = 0 \dots \text{Either } P(A) = 0 \text{ or } P(B) = 0 \text{ or both}$$

Hence two events cannot be both independent & mutually exclusive simultaneously with non-zero probabilities.

6) The weight of football players is normally distributed with a mean of 200 pounds & a std. deviation of 25 pounds. The probability of a player weighing more than 241.25 pounds is

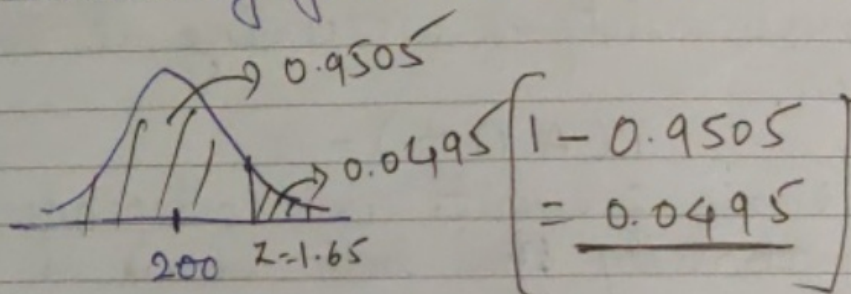
solⁿ

$$\mu = 200, \sigma = 25$$

$$Z\text{-score} = \frac{x - \text{mean}}{\sigma} = \frac{241.25 - 200}{25}$$

$$= 1.65$$

Using normal distribution table, the probability of a z-score being greater than 1.65



$$\begin{array}{r} 1.0000 \\ - 0.9505 \\ \hline 0.0495 \end{array}$$

$$\begin{aligned} P(x > 241.25) &= P(z > 1.65) \\ &= 1 - P(z < 1.65) \\ &= 1 - 0.9505 \\ &= 0.0495 \end{aligned}$$

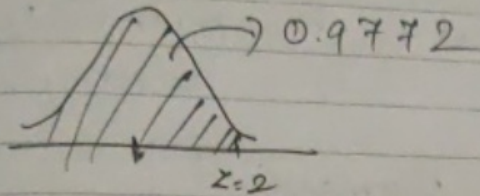
7) Refer to Q. 6, The probability of player weighing less than 250 pounds is

$$\mu = 200, \sigma = 25$$

$$Z\text{-score} = \frac{x - \text{mean}}{\sigma} = \frac{250 - 200}{25}$$

Using normal distribution table, the probability of z-score less than 2 is

$$P(X < 250) = P(Z < 2) \\ = 0.9772$$



9) Assuming a binomial experiment with $p=0.5$ & a sample size of 100. The expected value of this distribution is.

$$\text{Mean} = E(X) = np \\ = 100 \times 0.5 = 50$$

$$\text{Variance} = V(X) = npq$$

Note:- Binomial distribution - discrete probability distribution.

$$P(X=x) = {}^nC_x p^x q^{n-x} \\ \text{where } p+q=1$$

- Q. A company has 140 employees, of which 30 are supervisors. Eighty of the employees are married & 20% of married employees are supervisors. If a company employee is randomly selected, what is the probability that the employee is married & is a supervisor?

	Married	Unmarried	
Supervisor	16	14	30
Non-supervisor	64	46	110
	80	60	140

The probability that the employee is married & is a supervisor

$$P(M \cap S) = \frac{16}{140} = 0.1143$$

- Q. Of the cars in a used car lot, 70% have AC & 40% have a CD player. 20% of the cars have both. What is the probability that a car has a CD player, given that it has AC - $P(CD|AC)$

	AC	No AC	
CD	.2	.2	.4
No CD	.5	.1	.6
	.7	.3	1

$$P(CD|AC) = \frac{P(CD \cap AC)}{P(AC)} = \frac{.2}{.7} = .2857$$

Problem on week 2

Q. Machines A, B & C all produce same 2 parts, X & Y. Of all the parts produced, m/c A produces 60%, machine B produces 30% & machine C produces 10%. In addition,

- 40% of the parts made by m/c A are part X.
- 50% of the parts made by m/c B are part X.
- 70% of the parts made by m/c C are part X.

A part produced by this company is randomly sampled & is determined to be an X-part. With the knowledge that it is an X part, revise the probabilities that the part came from 'm/c A, B or C

	A	B	C	
X	.24	.15	.07	.46
Y	.36	.15	.03	.54
	.6	.3	.1	1

$$P(A|X) = \frac{P(A \cap X)}{P(X)} = \frac{.24}{.46} = 0.52$$

$$P(B|X) = \frac{P(B \cap X)}{P(X)} = \frac{.15}{.46} = .33$$

$$P(C|X) = \frac{P(C \cap X)}{P(X)} = \frac{.07}{.46} = 0.15$$

Q. A particular type of printer ribbon is produced by only 2 companies, Hama Ribbon Company & South Jersey products. Suppose Hama produces 65% of the ribbons & the South Jersey produces 35%. Eight percent of the ribbons produced by Hama are defective & 12% of South Jersey ribbons are defective. A customer purchases a new ribbon. What is the

probability that alamo produced the ribbon?
what is the probability that the South terry
produced the ribbon?

Q. 1

Q.

- Q. Use the data below to find out the expected number of credit cards that a customer to a retail outlet will possess.
 $x = \#$ credit cards.

x	$P(x=X)$
0	0.08
1	0.28
2	0.38
3	0.16
4	0.06
5	0.03
6	0.01

$$\begin{aligned}
 E(X) &= x_1 p_1 + x_2 p_2 + \dots + x_n p_n \\
 &= 0(0.08) + 1(0.28) + 2(0.38) + 3(0.16) + 4(0.06) + 5(0.03) + 6(0.01) \\
 &= 1.97 \\
 &\text{about 2 credit cards.}
 \end{aligned}$$

- Q. The quiz scores for a particular student are given below.

22, 25, 20, 18, 12, 20, 24, 20, 20, 25, 24, 25, 18.

Find the variance & std. Deviation

value	12	18	20	22	24	25
frequency	1	2	4	1	2	3
probability	.08	.15	.31	.08	.15	.23

$$\mu = 21$$

$$V(x) = p_1 (x_1 - \mu)^2 + p_2 (x_2 - \mu)^2 + \dots + p_n (x_n - \mu)^2$$

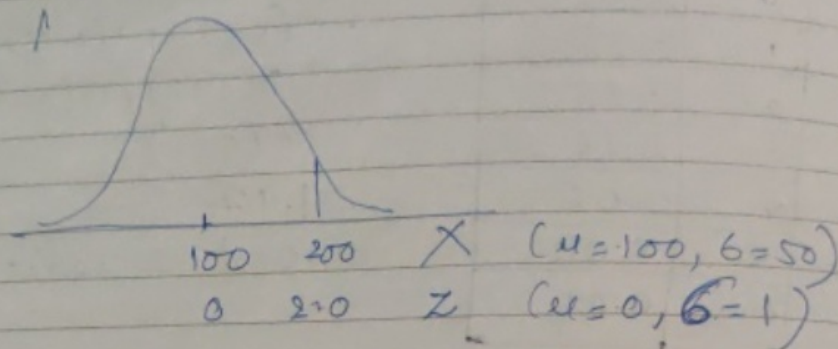
$$\sigma = \sqrt{V(X)}$$

$$\begin{aligned}
 V(x) &= .08(12-21)^2 + .15(18-21)^2 + .31(20-21)^2 + \\
 &\quad .08(22-21)^2 + .15(24-21)^2 + .23(25-21)^2 \\
 &= V(X) = 18.25
 \end{aligned}$$

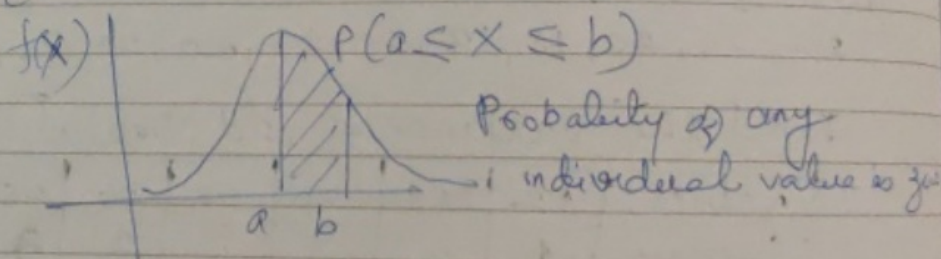
$$\sigma = \sqrt{V(X)} = \sqrt{18.25} = 3.64$$

Q. If X is distributed normally with mean of 100 & std. deviation of 50, the z value for $X=200$ is

$$Z = \frac{X - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$



Probability is measured by the area under the curve



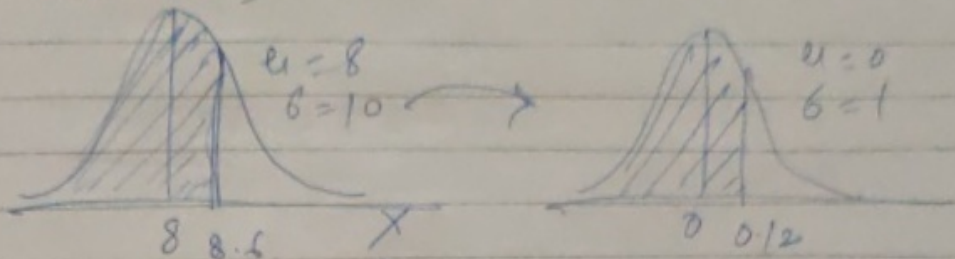
Finding Normal probability procedure

To find $P(a < X < b)$

- 1) Draw the normal curve for the problem in terms of X
- 2) Translate X values to Z values
- 3) Use the standard normal table

Q Let X represent the time it takes to download an image file from the internet. Suppose X is normal with mean 8.0 and std. deviation 5.0.

Find $P(X < 8.6)$



convert X to z -value

$$Z = \frac{X - \mu}{\sigma} = 0.12$$

Q. If the true proportion of voters who support proposition A is $p = .4$. What is the probability that a sample size of 200 yields a sample proportion between .40 & .45

$$p = .4, n = 200$$

$$P(.40 \leq \hat{p} \leq .45)$$

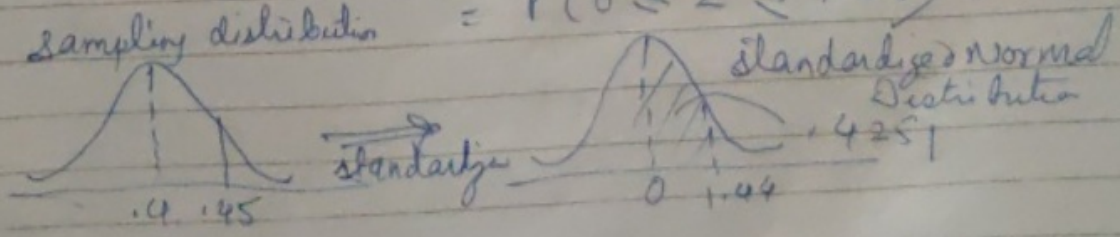
Ans:- find $\sigma_{\hat{p}}$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{.4(1-.4)}{200}} = 0.03464$$

convert to std normal.

$$P(.40 \leq \hat{p} \leq .45) = P\left(\frac{.40 - .40}{.03464} \leq Z \leq \frac{.45 - .40}{.03464}\right)$$

$$= P(0 \leq Z \leq 1.44) = .4251$$



Q. A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population std deviation is 0.35 ohms. Determine a 95% confidence interval for the true mean resistance of the population.

$$n = 11, \bar{x} = 2.20, \sigma = 0.35$$

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

5% - significance level - type I error.
- producer's risk.

Q. A random sample of $n = 25$ has $\bar{x} = 50.2$ and $s = 8$. Form a 95% confidence interval for μ .
d.f. = $n - 1 = 24$.

$$t_{n-1, \alpha/2} = t_{24, 0.025} = 2.0639$$

The confidence interval is

$$\bar{x} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

$$50.2 - 2.0639 \frac{8}{\sqrt{25}} < \mu < 50.2 + (2.0639) \frac{8}{\sqrt{25}}$$

$$46.698 < \mu < 53.802$$

Week 3 - Assignment 3

- 3) If the true proportion of customers who are below 20 years is $p = 0.35$. What is the probability that a sample size of 100 yields a sample proportion between 0.3 to 0.4.

soln $P(0.3 \leq \hat{p} \leq 0.4) \quad p = 0.35$
standardize $\hat{p}$ to 'z'

$$\text{Find } \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.35)(0.65)}{100}} = \frac{0.048}{0.4769}$$

$$P(0.3 \leq \hat{p} \leq 0.4) = P\left(\frac{0.3 - 0.35}{0.048} \leq z \leq \frac{0.4 - 0.35}{0.048}\right)$$

$$= P(-0.1042 \leq z \leq 0.2097)$$

$$0.4602 = P\left(\frac{-0.05}{0.048} \leq z \leq \frac{0.05}{0.048}\right)$$

$$= P(-1.042 \leq z \leq 1.042)$$

$$.1492 \quad .8508$$

$$.8508 - .1492 = .7016$$

$$= \underline{\underline{0.7016}}$$

- 4) Calculate the probability of getting 12 heads in 20 attempts from a fair coin.
 $n = 20, X = 12$

$$P(X=12) = {}^{20}C_{12} p^{12} q^{n-12}$$

$$= {}^{20}C_{12} (0.5)^{12} (0.5)^8$$

$$\left| \begin{array}{l} p = 0.5 \\ q = 0.5 \end{array} \right.$$

$${}^{20}_C_{12} = \frac{20!}{12! \cdot 8!} = \frac{20 \times 19 \times 18 \times 17 \times 16 \times 15 \times 14 \times 13 \times 12!}{12! \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= 5 \times 19 \times 3 \times 17 \times 2 \times 13$$

$$= 125970$$

$$125970 \times 0.00024 \times 0.0039 = 0.12$$

In Python

```
print(binom.pmf(K=12, n=20, p=0.5))
; 0.12018435363769531
```

Week 3 (extra problems)

Q. Suppose a large population has mean $\mu = 8$, & std deviation $\sigma = 3$. Suppose a random sample of size $n = 36$ is selected. What is the probability that sample mean is betⁿ 7.8 & 8.2

Solⁿ Even if the population is not normally distributed but the central limit theorem can be used ($n > 25$)

$$\therefore \mu_{\bar{x}} = 8$$

standard deviation of sampling distribution

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{36}} = \frac{3}{6} = .5$$

$$P(7.8 < \bar{x} < 8.2) = P\left[\frac{7.8 - 8}{3/\sqrt{36}} < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < \frac{8.2 - 8}{3/\sqrt{36}}\right]$$

$$= P(-0.5 < Z < 0.5) = 0.3830$$

from NPTEL

57 A question paper contains 90 multiple-choice questions. Each correct answer carries 1 mark. There are 4 alternative answers (A, B, C or D), of which only one is correct. Mr. X answers these questions randomly (i.e. without preparation). What is the probability that X gets at least 10 marks.

Python :- `1 - binom.cdf(K=9, n=90, p=0.25)`

- 6) On average 5% of items supplied by manufacturer are defective. If a batch of 10 items is inspected what is the probability that 2 items are defective

Soln

probability of defective item = 0.05
 " not defective = 0.95

$$K = 2, n = 10$$

$$\begin{aligned} P(X=2) &= {}^{10}C_2 \cdot (0.05)^2 \cdot (0.95)^8 \\ &= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2! \times 8!} \times 0.0025 \times 0.66 \\ &= 45 \times 0.0025 \times 0.66 \\ &= 0.07425 \end{aligned}$$

- 7) A car distributor in city Y experience an average of 2.5 car sales daily. Find the probability that on a randomly selected day, they will sell 5 cars.

$$\lambda = 2.5, x = 5$$

$$P(X=5) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{(2.5)^5 \cdot e^{-2.5}}{5!}$$

$$\frac{97.66 \times 0.082}{120}$$

- 8) Find the probability that on a randomly selected day, they will sell no cars.

$$\lambda = 2.5, x = 0$$

$$\begin{aligned} P(X=0) &= \frac{\lambda^x e^{-\lambda}}{x!} = \frac{(2.5)^0 e^{-2.5}}{0!} = e^{-2.5} \\ &= 0.082 \end{aligned}$$

9) In question 7, find the probability that on a randomly selected day, they will sell at most 2 cars

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$P(X=1) = \frac{(2.5)^1 e^{-2.5}}{1!} = 2.5 \times 0.082 = 0.205$$

$$P(X=2) = \frac{(2.5)^2 e^{-2.5}}{2!} = \frac{2.5 \times 2.5 \times 0.082}{2} = 0.256$$

$$P(X \leq 2) = 0.082 + 0.205 + 0.256 = 0.543$$

10) In question 7, find the probability that on a randomly selected day, they will sell exactly one car.

$$P(X=1) = 0.205 \text{ (As calculated in q. 9)}$$

Week 4 - Extra problems

if σ known

The mean response times for a random sample of 30 pizza deliveries is 32 minutes. The population std deviation is believed to be 10 min. The pizza delivery services director wants to perform a hypothesis test with $\alpha = 0.05$ level of significance, to determine whether the service goal of 30 min or less is being achieved.

Given

Sample mean = 32 min ($\bar{x}$)

sample size = 30 (n)

population

$\alpha = 0.05$

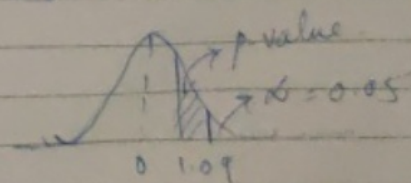
population mean = 30 min

$\sigma = 10$ min.

p-value approach

1. Develop the hypotheses $H_0 - \mu \leq 30$
2. Specify level of significance. $H_a - \mu > 30$
3. Compute the value of test statistic $\alpha = 0.05$

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{32 - 30}{10 / \sqrt{30}} = 1.09$$



4. Compute the p-value
for $Z = 1.09$, $p\text{-value} = 0.137$

5. Determine whether to reject H_0 .

As $p\text{-value} = 0.137 > \alpha$, we do not reject H_0 .

$\Rightarrow$ There are not sufficient statistical evidence to infer that pizza delivery services is not meeting the response goal of 30 min.

Critical value approach

Step 1 to 3 are same as above

4) Determine critical value & rejection rule

$$\text{for } \alpha = .05, z_{.05} = 1.645$$

Reject H_0 if $z \geq 1.645$.

5) Determine whether to reject H_0

- Because $1.645 > 1.05$, we do not reject H_0

Week 4 problems (NPTEL)

Q.384) The quality-control manager at Li-Battery factory needs to determine whether the mean life of a large shipment of Li-battery is equal to the specified value of 375 hours. The process std. deviation is known to be 100 hours. A random sample of 64 batteries indicates a sample mean life of 350 hrs. State the null hypothesis. At the $\alpha = 0.05$ level of significance, is there any evidence that mean life differs from 375 hours.

p-value
to be

Soln

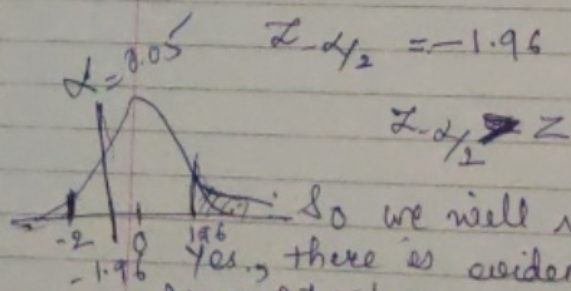
$$\mu = 375, \sigma = 100.$$

$$\bar{x} = 350, n = 64.$$

Null hypothesis: $\mu = 375$

Alternate hypothesis: $\mu \neq 375$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{350 - 375}{100/\sqrt{64}} = \frac{-25}{100/8} = -2$$



So we will reject null hypothesis
Yes, there is evidence that mean life is different from 375 hrs.

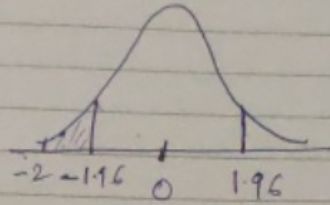
5) In above question p-value is

For $Z = -2$, $p\text{-value} = 0.01831$

$$P(Z < -2) = 1 - 0.98169$$

attorney factory
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p-value - It is the probability for the null hypothesis to be true



-2

different